Modular Athletic Vest

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Project Justification

- Most fitness wearables (Garmin, Fitbit, WHOOP) operate in **closed ecosystems**
- Users can only connect a **limited set of approved sensors**
- Advanced metrics (muscle activation, force, custom biomechanics) are **not supported**

Project Description

The Modular Athletic Vest is an Arduino-based wearable designed to provide **standard fitness tracking:**

- Heart rate monitoring
- Step counting
- Speed tracking
- Real-time exercise feedback based on user goals
- Workout data storage on a microSD card

Unlike traditional wearables, the vest has a **modular architecture**, allowing users to integrate **additional Arduino-compatible sensors** to measure advanced physiological or biomechanical metrics.

Target Users

- Coaches analyzing athlete performance
- Competitive athletes needing advanced training metrics
- Tech-savvy fitness enthusiasts interested in custom hardware

Design Process

The PCB layout includes connections for the Arduino R4, MPU6050 IMU, GPS module, OpenLog SD storage, and LCD interface. The design incorporates standardized mounting holes, optimized trace routing, and JST connectors for reliable communication and power distribution across the system (Fig. 1).

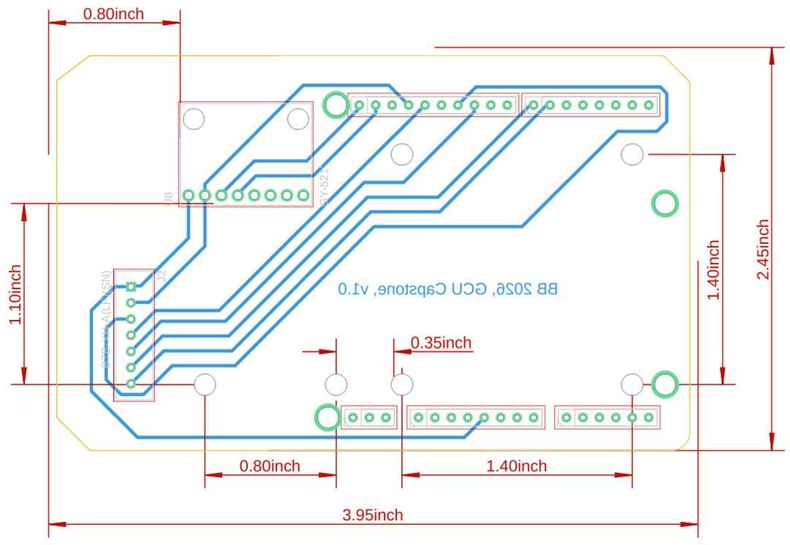

Figure 1. Custom PCB layout for an integrated sensor and microcontroller system.

System workflow from setup to output includes initialization, sensor integration, and data processing. Real-time inputs (heart rate, steps, and speed) are filtered and processed before being displayed on the LCD and stored for later analysis (Fig. 2).

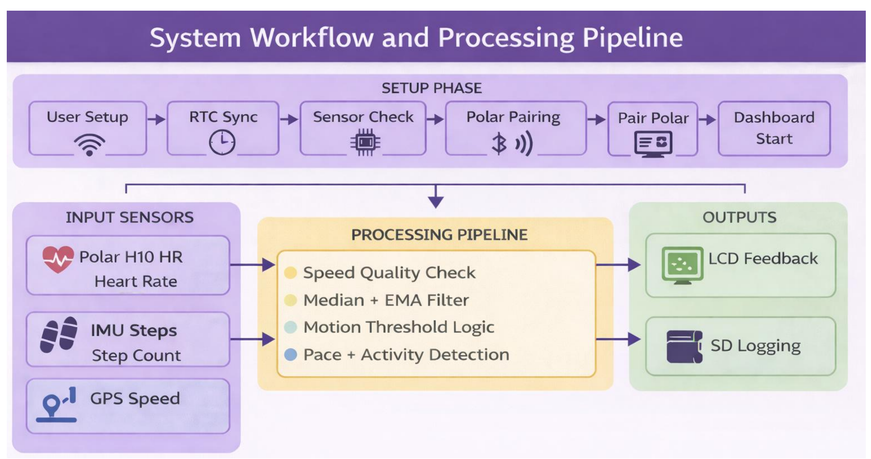

Figure 2. System workflow and real-time processing pipeline for the modular athletic vest.

A three-part enclosure houses the Arduino-PCB system and battery. The vest shows enclosure integration into the lower compartment with wiring for the LCD interface that was Velcro-mounted on a wrist case (Fig. 3).

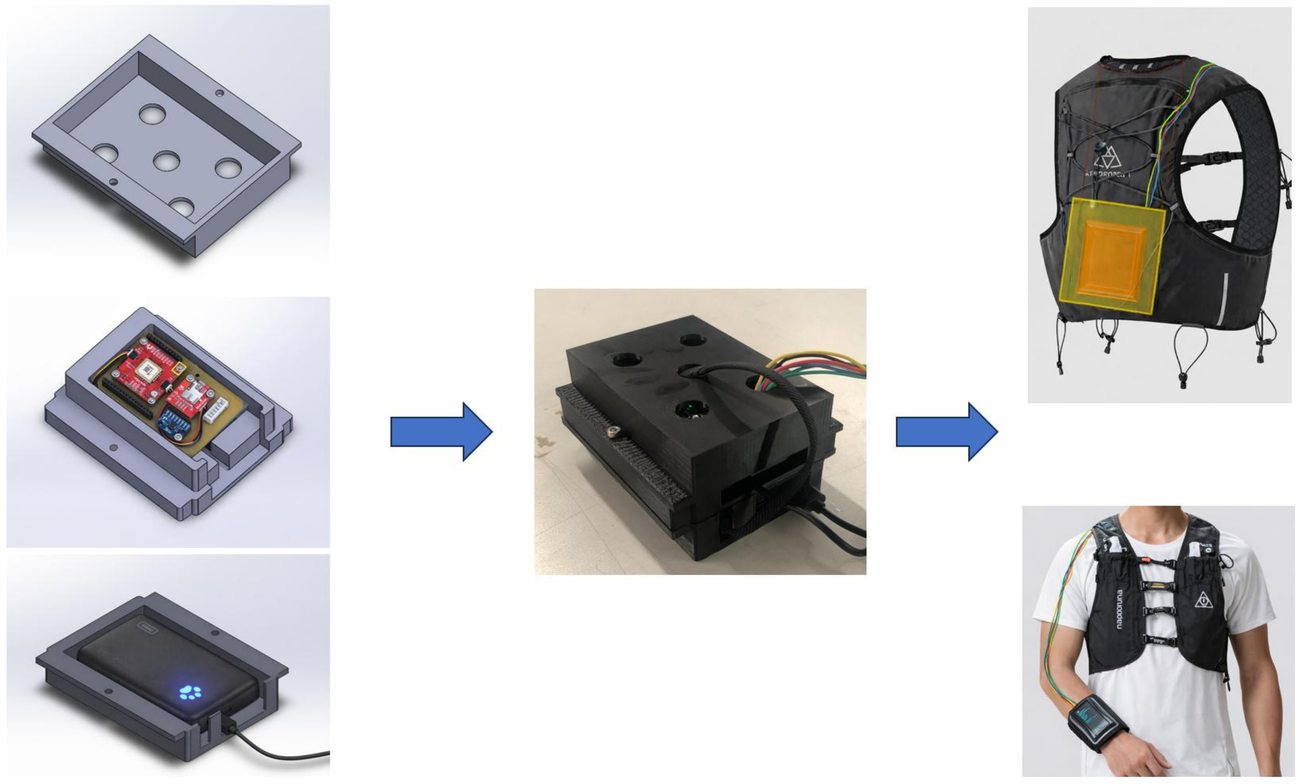

Figure 3. Enclosure design and integration into the wearable vest system.

System Performance & Validation

The functional requirements assessment confirms that the modular athletic vest meets all key performance criteria, including accuracy in heart rate (100% specificity), step tracking (MAPE = 4.69%), speed tracking (MAPE = 8.86%), battery life, data storage, and enclosure design. Validation was guided by established standards such as IEC 62133-2:2017, IEC 60601-2-47:2012, ISO 9241-210:2019, and ISO 12100, with additional benchmarking methods used where formal standards were not available.

Functional Requirement	Product Specification	Engineering Standards/ Alternative Tests	Result (Pass/Fail)
Long battery life	Runtime > 24 hours	IEC 62133-2:2017	**FAIL (23.5 hours)**
Accurate heart rate detection	Specificity > 91%	IEC 60601-2-47:2012	**PASS (100%)**
Accurate step tracking	MAPE < 20% (Mora-Gonzalez et al., 2022)	Manual step count comparison	**PASS (4.69%)**
Accurate speed tracking	MAPE < 10% (Alphin et al., 2020)	Radar gun comparison	**PASS (8.86%)**
Effective user feedback system	Rating ≥ 7/10	ISO 9241-210:2019	**PASS (8.69)**
User comfort	Rating ≥ 7/10	ISO 9241-210:2019	**PASS (7.82)**
Reliable data storage	Success rate ≥ 90%	Functional storage test	**PASS (100%)**
Modular enclosure	≥ 2 holes ≥ 15 mm	Dimensional measurement	**PASS**
Secure enclosure locking	No failure under use	ISO 12100	**PASS**

References

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- Alphin, K. L., Sisson, O. M., Hudgins, B. L., Noonan, C. D., & Bunn, J. A. (2020). Accuracy assessment of a GPS device for maximum sprint speed. *International Journal of Exercise Science, 13*(4), 273–280.
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